

Title:

Hidden Data Leakage in OCT-Based Medical Imaging AI: A Systematic Framework for Detection, Mitigation, and Leakage-Aware Evaluation

Abstract

Hidden data leakage can substantially bias performance estimates in medical imaging artificial intelligence (AI), yet its impact is not routinely quantified. Using retinal optical coherence tomography (OCT) classification as a representative use case, we developed and evaluated a leakage-aware framework to detect and mitigate hidden data leakage. Three dataset configurations were constructed from OCT2017 with progressively stricter leakage control: the original leakage-prone dataset, a duplication-controlled dataset generated by removing exact duplicate images using SHA-256 hashing, and a fully leakage-controlled dataset further refined by eliminating patient-level overlap using GroupKFold-based subject separation. Seven deep learning architectures were trained under a unified protocol, and performance was assessed using macro-averaged AUROC, macro-F1, recall, specificity, precision, and negative predictive value. External validation was performed on OCT-C8 and LOCTv3, and statistical comparisons were conducted using bootstrap resampling with false-discovery-rate correction. Internal performance remained high across all dataset configurations. Removal of exact duplicate images alone had minimal impact, whereas strict patient-level separation consistently reduced performance across models, with larger effects on macro-F1 than AUROC. Models trained on the fully leakage-controlled dataset showed smaller generalization gaps on LOCTv3, suggesting improved robustness under distributional shift. These findings indicate that patient-level leakage is a major source of hidden performance inflation in OCT classification.

Keywords:

optical coherence tomography; data leakage; deep learning; external validation; generalization; medical imaging AI

1 Introduction

Artificial intelligence (AI), particularly deep learning, has shown substantial promise in the analysis of retinal optical coherence tomography (OCT) images. Prior studies have reported strong diagnostic performance for automated classification of retinal conditions, including choroidal neovascularization (CNV), diabetic macular edema (DME), drusen, and normal retina, with area under the receiver operating characteristic curve (AUROC) values often approaching or exceeding 0.99¹⁻⁵. While these findings suggest that AI systems may approach expert-level performance, concerns remain regarding the validity and generalizability of such results.

One critical but often underrecognized issue is data leakage in publicly available datasets. Widely used OCT datasets, such as OCT2017, have been reported to contain duplicated images as well as potential patient-level overlap across training, validation, and test partitions^{6,7}. These violations of data independence can lead to overly optimistic performance estimates, thereby undermining the reliability of model evaluation. Despite its potential impact, data leakage is not consistently addressed or systematically quantified in prior OCT classification studies. External validation is commonly used to assess model generalizability⁸, but its reliability depends on the assumption that external datasets are independent of the training data⁹. In practice, this assumption is difficult to verify when publicly available datasets are used. External test sets may share common sources, contain redistributed subsets, or include overlapping patient populations that are not explicitly documented¹⁰. Consequently, external validation results may still be affected by hidden data leakage, potentially

leading to misinterpretation of model robustness. Importantly, OCT imaging datasets possess structural characteristics that enable systematic detection and mitigation of certain forms of data leakage. Exact duplicate images can be identified through file-level analysis, while patient-level overlap can be assessed using identifiers embedded in image metadata or file naming conventions. These characteristics provide an opportunity to establish reproducible and scalable data curation strategies aimed at improving dataset integrity.

In this study, we propose a leakage-aware evaluation framework for OCT classification that explicitly accounts for both file-level and patient-level sources of data leakage. Three dataset configurations with progressively stricter leakage control were constructed, including the original dataset, a duplication-controlled dataset, and a fully leakage-controlled dataset with patient-level separation. Using these datasets, we systematically evaluated the impact of data leakage on model performance across multiple architectures. In addition, external validation was conducted on independent datasets, and generalization gaps were quantified to assess robustness under distributional shift.

This work aims to provide a more rigorous and clinically relevant assessment of AI performance in OCT classification and to highlight the importance of leakage-aware evaluation in medical imaging research.

2 Materials and Methods

2.1 Ethics Statement

This study used only publicly available, fully anonymized datasets and involved no identifiable human information. Therefore, Institutional Review Board approval and informed consent were not required under applicable regulations.

2.2 Study Design and Overall Pipeline

The overall study workflow is summarized in Fig. 1. This study evaluated the impact

of hidden data leakage on OCT classification under progressively stricter leakage-control conditions. The workflow consisted of four stages: (1) dataset collection and categorization, (2) dataset curation and leakage control, (3) model development and internal validation, and (4) external validation and generalization analysis. First, three publicly available OCT datasets were used: OCT2017¹¹, OCT-C8¹², and LOCTv3¹³. OCT2017 served as the primary dataset for model development and internal validation, whereas OCT-C8 and LOCTv3 were reserved exclusively for external validation. Second, OCT2017 was progressively reconstructed into three dataset versions with increasing levels of leakage control: the original leakage-prone dataset, a duplication-controlled dataset, and a fully leakage-controlled dataset with strict patient-level separation. Models were trained and internally evaluated under a unified protocol, and the best-performing models were externally evaluated on OCT-C8 and LOCTv3 to quantify generalization gaps.

2.3 Datasets

OCT2017 served as the basis for leakage-aware dataset reconstruction¹¹. It comprises 84,484 OCT B-scan images labeled into four diagnostic categories: choroidal neovascularization (CNV), diabetic macular edema (DME), drusen, and normal. Because OCT2017 has been reported to contain duplicated images and potential patient-level overlap between training and testing partitions, it was selected for leakage-control reconstruction.

OCT2017 was further organized into three versions with increasing levels of leakage control: the original leakage-prone dataset, a duplication-controlled dataset (Ravelli-derived), and a fully leakage-controlled dataset (OCT-revised) with strict patient-level separation. These versions shared identical class labels and enabled controlled comparison across different leakage-control conditions, with OCT-revised used as the primary reference for performance comparison.

OCT-C8¹² and LOCTv3¹³ each contained 1,000 test images and were not

involved in model development, training, or model selection. Detailed statistics of duplicate images and patient-level overlap are provided in Supplementary Fig. S1.

2.4 Dataset Cleaning and Leakage Control

To address data leakage, we implemented a two-stage framework consisting of file-level duplicate detection and subject-level overlap elimination. Data leakage in medical imaging datasets can be broadly categorized into three forms: (1) exact duplication (file-level redundancy) ¹⁴, (2) near-duplicate or slice-level similarity ^{7,14}, and (3) subject-level overlap across dataset partitions ¹⁵. In this study, we explicitly addressed the first and third forms, which are most prevalent in publicly available OCT datasets.

First, file-level duplicate removal was performed using the SHA-256 cryptographic hashing algorithm ¹⁶. Each image file was processed in binary format and converted into a unique 256-bit hash signature representing its exact binary content. Images sharing identical hash values were considered exact duplicates. Using this approach, 7,877 duplicated images were identified and removed from the OCT2017 dataset, yielding the Ravelli-derived dataset. Second, patient-level overlap elimination was performed to prevent data leakage caused by images from the same patient appearing in different dataset splits ^{7,17}. Patient identifiers were extracted directly from the file naming convention [class]-[patientID]-[scanID]. For example, CNV-53018-1.jpeg denotes the disease class CNV, patient ID 53018, and scan index 1. Using these identifiers, images from the same patient were grouped and partitioned using GroupKFold ¹⁸, ensuring that all images from a given patient were assigned to the same data partition. With this approach, 7,134 cross-partition patient-level overlaps were identified and eliminated, yielding the final OCT-revised dataset with strict subject-level independence. Class distribution and leakage statistics are summarized in Tables 1 and 2.

2.5 Image Preprocessing

All OCT images were resized to 224×224 pixels, converted to tensor format, and normalized using ImageNet statistics: mean = [0.485, 0.456, 0.406] and standard deviation = [0.229, 0.224, 0.225]. This normalization aligns the input distribution with that expected by ImageNet-pretrained models¹⁹ and facilitates stable optimization during training.

Data augmentation was applied only to the training set and included horizontal flipping with a probability of 0.5, random rotation within $\pm 10^\circ$, and random cropping with a scale range of 0.8–1.0. No augmentation was applied to the validation or test sets to ensure unbiased performance evaluation.

2.6 Model Architectures

Seven deep learning architectures were evaluated, including convolutional neural networks (CNNs), vision transformers, and hybrid architectures. The CNN-based models included ResNet-50²⁰, EfficientNet-V2-M²¹, DenseNet-169, and ConvNeXt-Tiny²². The transformer-based architecture was Swin Transformer (Swin-T)²³. The hybrid architectures included CoAtNet-2²⁴ and MaxViT-Base²⁵.

All models were initialized with ImageNet-pretrained weights (IMAGENET1K_V2) and adapted for four-class OCT classification. Most models were implemented in PyTorch, whereas CoAtNet was implemented using the timm library.

2.7 Training Strategy and Hyperparameters

All experiments were conducted on a workstation equipped with an NVIDIA RTX 4090 GPU (24 GB VRAM), using Python 3.11.14 and PyTorch 2.10.0. Models were optimized using AdamW with a learning rate of 3×10^{-4} , batch size of 32, and 20 training epochs. A weight decay of 1×10^{-4} was used for regularization. A cosine annealing learning-rate scheduler was applied during training to gradually reduce the

learning rate (CosineAnnealingLR, $T_{\text{max}} = \text{epochs}$). The training objective was class-weighted cross-entropy loss to account for class imbalance in the dataset. Early stopping was not used.

StratifiedKFold ($K = 5$) was used for the OCT2017 and Ravelli-derived datasets to preserve class distribution across folds, whereas GroupKFold ($K = 5$) was used for the OCT-revised dataset to ensure strict patient-level separation and prevent subject-level data leakage.

2.8 Performance Evaluation

Model performance was evaluated using standard quantitative metrics for medical image classification, including accuracy [Eq. S1], sensitivity (recall) [Eq. S2], specificity [Eq. S3], precision (PPV) [Eq. S4], negative predictive value (NPV) [Eq. S5], F1-score [Eq. S6], and area under the receiver operating characteristic curve (AUROC). All metrics were reported using macro-averaging. Confusion matrices and ROC curves were also generated.

2.9 Statistical Analysis

Statistical analyses were performed to determine whether observed performance differences between datasets and models were statistically significant. Normality of fold-wise performance distributions was first assessed using the Shapiro–Wilk test. Because the assumption of normality was not consistently satisfied, nonparametric approaches were adopted. Performance differences between datasets were quantified using a bootstrap resampling framework. For each comparison, metric differences were estimated using 2,000 bootstrap iterations. This procedure was applied independently to each fold to obtain empirical distributions of performance differences. From these distributions, mean differences, 95% confidence intervals (CIs), and two-sided empirical p-values were derived. To summarize results across cross-validation folds, fold-level p-values were combined using Fisher’s method, and the mean and variability of the performance differences were reported. Multiple

comparisons were adjusted using the Benjamini–Hochberg procedure to control the false discovery rate (FDR) across all tested hypotheses.

All statistical analyses were based on macro-averaged area under the receiver operating characteristic curve (AUROC). In addition, macro-F1 score and class-specific sensitivity for DRUSEN were analyzed to provide complementary insights into classification performance, particularly for a clinically relevant minority class. A significance level of $\alpha = 0.05$ was used, and all statistical tests were two-sided. In addition to statistical significance, effect sizes and corresponding confidence intervals were reported to facilitate interpretation of practical relevance. Generalization gap was defined as the external minus internal performance ($\Delta = \text{external} - \text{internal}$).

3 Results

3.1 Overall Performance Across Datasets

Table 3 presents the overall performance comparison across three dataset configurations for OCT classification. All evaluated models showed consistently high internal performance across the three dataset configurations (OCT2017, Ravelli-derived, and OCT-revised). EfficientNet-V2-M achieved the highest AUROC (0.9929 ± 0.0011) on the OCT-revised dataset. Fig. 4 further illustrates class-specific performance across the three dataset configurations for EfficientNet-V2-M. As shown in Fig. 4, macro-averaged AUROC exceeded 0.97 for all architectures, while macro-F1 remained above 0.95 in OCT2017 and Ravelli-derived. These findings indicate near-ceiling performance under conventional evaluation settings [31, 36], but performance declined systematically as leakage control became more stringent. Differences between OCT2017 and Ravelli-derived were minimal, whereas a consistent reduction was observed in OCT-revised across all models. Notably, the decrease was more pronounced in macro-F1 compared to AUROC, suggesting that threshold-dependent metrics are more sensitive to leakage removal than ranking-based metrics.

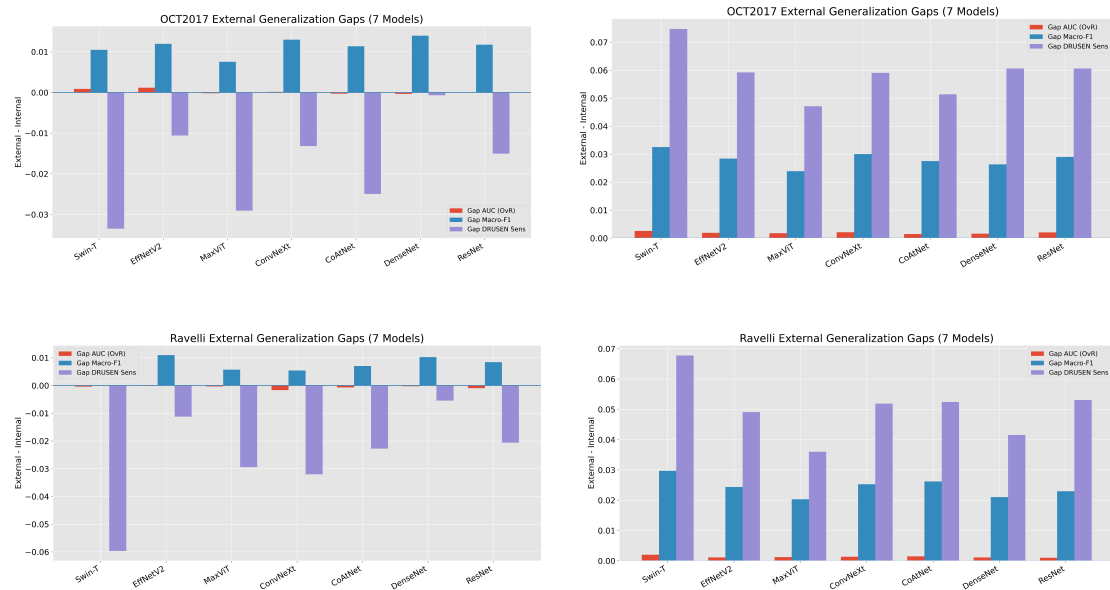
The negligible differences between OCT2017 and Ravelli-derived, despite the removal of 7,877 duplicated images, suggest that file-level duplication alone does not substantially inflate model performance. In contrast, the transition from Ravelli-derived to OCT-revised resulted in a consistent performance drop across all architectures, indicating that patient-level leakage is a major source of hidden performance inflation in OCT classification.

3.2 Model Behaviour Analysis

Fig. 5 shows the distribution of DRUSEN-to-NORMAL misclassification rates across folds. A clear upward shift in the median error rate is observed from OCT2017 to OCT-revised, indicating that stricter leakage control increases misclassification in this clinically challenging category. In addition to the increase in median, the OCT-revised dataset exhibits substantially greater variability, with a wider interquartile range and the presence of high-error outliers. This suggests that performance degradation is not uniform, but instead concentrated in more difficult cases. By contrast, the Ravelli-derived dataset showed only minor changes compared to OCT2017, indicating that image-level duplicate removal alone has limited impact on model error patterns.

3.3 External Validation and Generalization Gap

External validation on the independent LOCTv3 dataset revealed substantial differences in generalization behavior across dataset configurations in Fig. 6a. Models trained on OCT2017 exhibited the largest generalization gaps, as evidenced by notable decreases in macro-F1 and class-specific sensitivity.



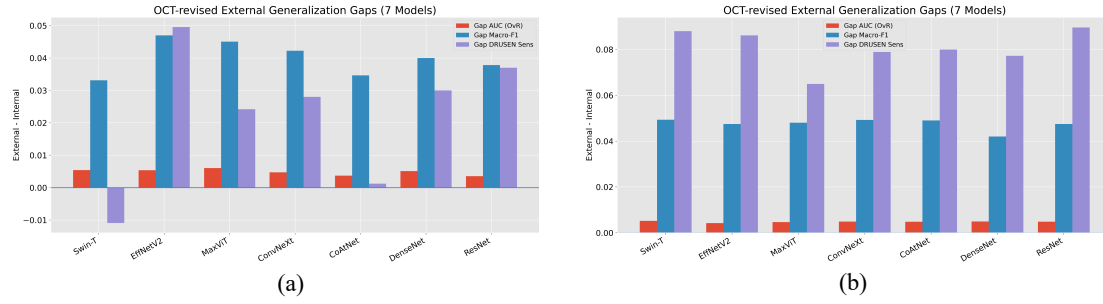


Figure 6. The figure shows performance differences (external – internal) for macro-AUROC, macro-F1, and DRUSEN sensitivity. On (a) LOCTv3, models trained on OCT2017 show the largest gaps, Ravelli shows moderate gaps, and OCT-revised shows the smallest, indicating improved robustness with stricter leakage control. On (b) OCT-C8, gaps are minimal or reversed, likely due to dataset overlap, highlighting the risk of hidden data leakage in external validation.

In comparison, models trained on the Ravelli dataset showed moderately reduced generalization gaps, whereas those trained on the OCT-revised dataset demonstrated the smallest discrepancies between internal and external performance. In several cases, the performance difference approached zero, indicating improved robustness to distributional shifts.

These findings suggest that stricter leakage control leads to more reliable estimates of real-world model performance.

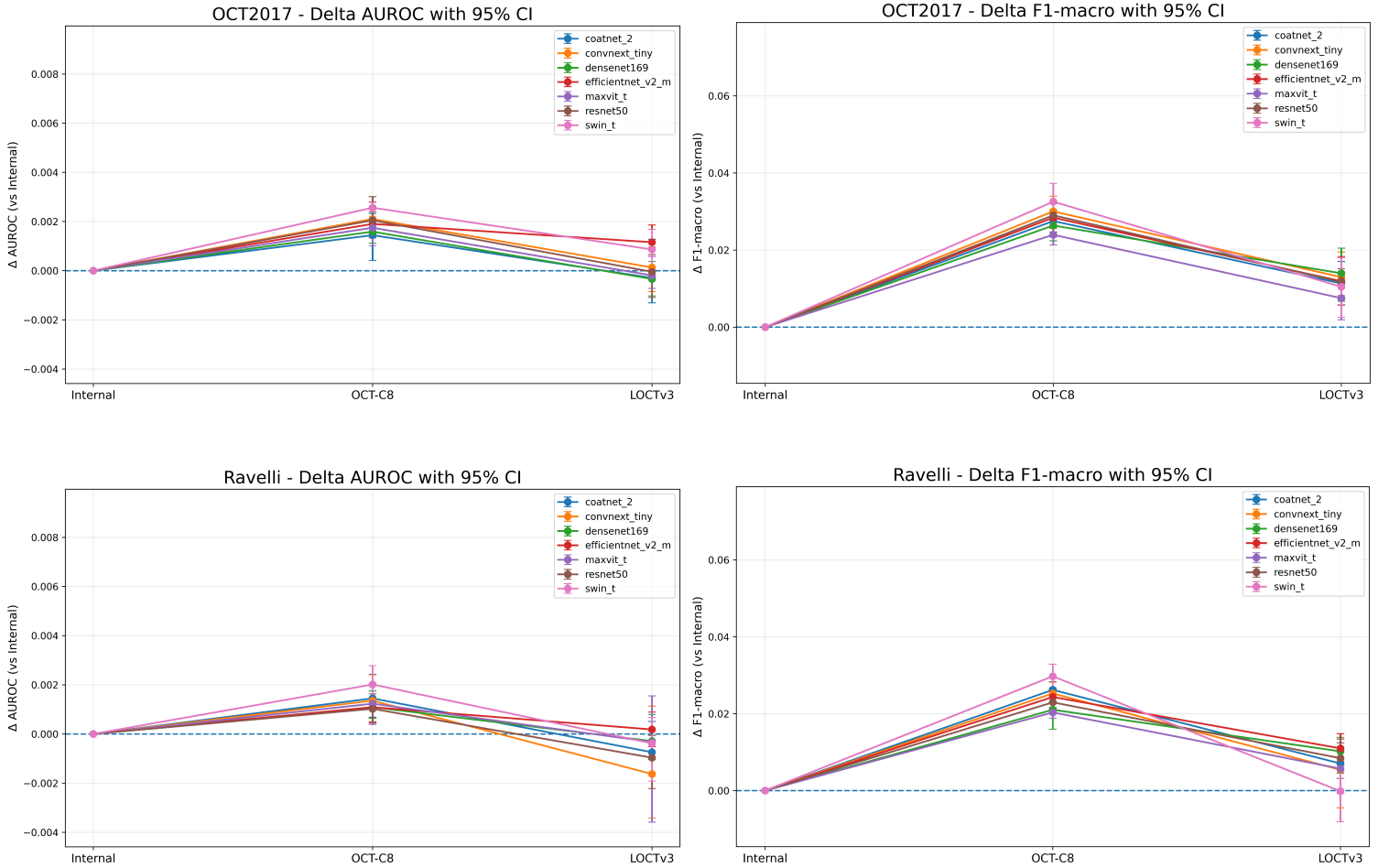
A contrasting pattern was observed on the OCT-C8 (Fig. 6b). Across all dataset configurations, models exhibited minimal or even reversed generalization gaps, with external performance occasionally exceeding internal performance.

Because OCT-C8 was later found to contain substantial overlap with the original OCT2017 dataset, these results indicate that apparent improvements in generalization may arise from hidden data leakage between training and external evaluation sets. This highlights the critical importance of verifying dataset independence in external validation studies.

3.4 Statistical Analysis

Bootstrap-based statistical analysis confirmed the observed performance trends across datasets. Pairwise comparisons of performance differences were conducted using a resampling framework [28], in which empirical distributions of metric differences were estimated for each fold.

Across models, performance differences between OCT2017 and Ravelli-derived were generally small, with confidence intervals frequently overlapping zero. Fig. 7a shows that duplicate removal alone had limited impact on model performance, and most comparisons did not demonstrate consistent or robust differences after multiple comparison correction.



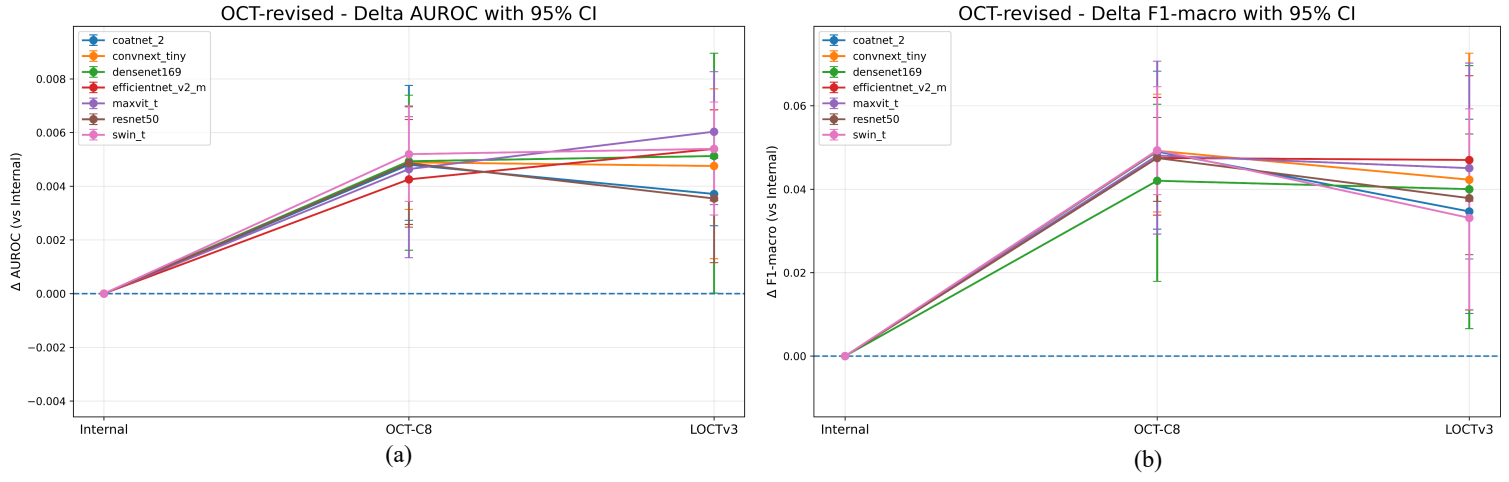


Figure. 7. The figure shows Δ AUROC (a) and Δ macro-F1 (b) with 95% confidence intervals. Differences between OCT2017 and Ravelli were small with confidence intervals overlapping zero, indicating limited impact of duplicate removal. In contrast, comparisons involving the OCT-revised dataset showed larger and more consistent shifts, especially for macro-F1. Although some AUROC differences were statistically significant, their magnitude remained small, suggesting limited practical impact, while macro-F1 was more sensitive to leakage removal.

In contrast, comparisons involving the OCT-revised dataset exhibited more consistent and larger performance differences, particularly for macro-F1 in Fig. 7b. These differences were reflected by confidence intervals that were more frequently shifted away from zero, suggesting a measurable impact of patient-level leakage removal on classification performance.

Dataset	Model	Comparison	Δ AUROC (External - Internal)	95%_CI_mean _delta_auroc
OCT2017	coatnet_2	Internal vs LOCTv3	-0.0003 $\pm$ 0.0011	[-0.001309, 0.001278]
		Internal vs OCT-C8	0.0014 $\pm$ 0.0008***	[0.000412, 0.002328]

		OCT-C8 vs LOCTv3	-0.0017 ± 0.0011	[-0.003070, - 0.000548]
	convnext_tiny	Internal vs LOCTv3	0.0001 ± 0.0008	[-0.000852, 0.000976]
		Internal vs OCT-C8	0.0021 ± 0.0005***	[0.001333, 0.002500]
		OCT-C8 vs LOCTv3	-0.0020 ± 0.0010*	[-0.003027, - 0.000877]
	densenet169	Internal vs LOCTv3	-0.0003 ± 0.0006	[-0.001026, 0.000372]
		Internal vs OCT-C8	0.0016 ± 0.0005***	[0.001122, 0.002385]
		OCT-C8 vs LOCTv3	-0.0019 ± 0.0007***	[-0.002571, - 0.000966]
	efficientnet_v2_m	Internal vs LOCTv3	0.0012 ± 0.0008**	[0.000009, 0.001859]
		Internal vs OCT-C8	0.0019 ± 0.0006***	[0.001344, 0.002789]
		OCT-C8 vs LOCTv3	-0.0007 ± 0.0007	[-0.001561, - 0.000154]
	maxvit_t	Internal vs LOCTv3	-0.0002 ± 0.0005	[-0.000721, 0.000577]
		Internal vs OCT-C8	0.0017 ± 0.0006***	[0.001018, 0.002619]
		OCT-C8 vs LOCTv3	-0.0019 ± 0.0005***	[-0.002248, - 0.001208]

	resnet50	Internal vs LOCTv3	0.0000 ± 0.0008	[-0.001088, 0.000653]
		Internal vs OCT-C8	0.0020 ± 0.0006***	[0.001520, 0.003011]
		OCT-C8 vs LOCTv3	-0.0021 ± 0.0006**	[-0.002782, - 0.001419]
	swin_t	Internal vs LOCTv3	0.0009 ± 0.0009	[-0.000388, 0.001676]
		Internal vs OCT-C8	0.0026 ± 0.0002***	[0.002403, 0.002774]
		OCT-C8 vs LOCTv3	-0.0017 ± 0.0008**	[-0.002790, - 0.000951]
Ravelli	coatnet_2	Internal vs LOCTv3	-0.0007 ± 0.0009	[-0.001587, 0.000509]
		Internal vs OCT-C8	0.0014 ± 0.0008***	[0.000680, 0.002427]
		OCT-C8 vs LOCTv3	-0.0022 ± 0.0013***	[-0.004014, - 0.000986]
	convnext_tiny	Internal vs LOCTv3	-0.0016 ± 0.0019	[-0.003416, 0.001134]
		Internal vs OCT-C8	0.0014 ± 0.0008***	[0.000643, 0.002404]
		OCT-C8 vs LOCTv3	-0.0030 ± 0.0012*	[-0.004108, - 0.001193]
	densenet169	Internal vs LOCTv3	-0.0003 ± 0.0008	[-0.001044, 0.000793]

		Internal vs OCT-C8	0.0011 $\pm$ 0.0005***	[0.000632, 0.001744]
		OCT-C8 vs LOCTv3	-0.0014 $\pm$ 0.0010	[-0.002731, - 0.000060]
	efficientnet_ v2_m	Internal vs LOCTv3	0.0002 $\pm$ 0.0006	[-0.000515, 0.000891]
		Internal vs OCT-C8	0.0011 $\pm$ 0.0007***	[0.000446, 0.002021]
		OCT-C8 vs LOCTv3	-0.0009 $\pm$ 0.0011	[-0.002487, 0.000142]
	maxvit_t	Internal vs LOCTv3	-0.0003 $\pm$ 0.0022	[-0.003586, 0.001547]
		Internal vs OCT-C8	0.0012 $\pm$ 0.0006**	[0.000384, 0.002045]
		OCT-C8 vs LOCTv3	-0.0016 $\pm$ 0.0021	[-0.004885, - 0.000197]
	resnet50	Internal vs LOCTv3	-0.0010 $\pm$ 0.0010	[-0.002224, - 0.000059]
		Internal vs OCT-C8	0.0010 $\pm$ 0.0006	[0.000500, 0.001628]
		OCT-C8 vs LOCTv3	-0.0020 $\pm$ 0.0007	[-0.002791, - 0.001127]
	swin_t	Internal vs LOCTv3	-0.0004 $\pm$ 0.0010	[-0.001916, 0.000668]
		Internal vs OCT-C8	0.0020 $\pm$ 0.0006***	[0.001477, 0.002776]

		OCT-C8 vs LOCTv3	-0.0024 $\pm$ 0.0007*	[-0.003446, - 0.001797]
OCT- revised	coatnet_2	Internal vs LOCTv3	0.0037 $\pm$ 0.0010***	[0.002529, 0.005041]
		Internal vs OCT-C8	0.0048 $\pm$ 0.0021***	[0.002729, 0.007754]
		OCT-C8 vs LOCTv3	-0.0011 $\pm$ 0.0024	[-0.004497, 0.001580]
	convnext_tiny	Internal vs LOCTv3	0.0048 $\pm$ 0.0025***	[0.001301, 0.007625]
		Internal vs OCT-C8	0.0049 $\pm$ 0.0016***	[0.003137, 0.006972]
		OCT-C8 vs LOCTv3	-0.0001 $\pm$ 0.0019	[-0.002078, 0.002154]
	densenet169	Internal vs LOCTv3	0.0051 $\pm$ 0.0036***	[0.000017, 0.008954]
		Internal vs OCT-C8	0.0049 $\pm$ 0.0024***	[0.001613, 0.007392]
		OCT-C8 vs LOCTv3	0.0002 $\pm$ 0.0015	[-0.001756, 0.001690]
	efficientnet_v2_m	Internal vs LOCTv3	0.0054 $\pm$ 0.0014***	[0.003454, 0.006846]
		Internal vs OCT-C8	0.0043 $\pm$ 0.0016***	[0.002577, 0.006481]
		OCT-C8 vs LOCTv3	0.0011 $\pm$ 0.0012	[-0.000167, 0.002781]

	maxvit_t	Internal vs LOCTv3	0.0060 ± 0.0022***	[0.003313, 0.008266]
		Internal vs OCT-C8	0.0046 ± 0.0022***	[0.001336, 0.006590]
		OCT-C8 vs LOCTv3	0.0014 ± 0.0010**	[0.000159, 0.002521]
	resnet50	Internal vs LOCTv3	0.0035 ± 0.0017***	[0.001153, 0.005130]
		Internal vs OCT-C8	0.0049 ± 0.0018***	[0.002477, 0.006956]
		OCT-C8 vs LOCTv3	-0.0013 ± 0.0008	[-0.002069, - 0.000325]
	swin_t	Internal vs LOCTv3	0.0054 ± 0.0019***	[0.002927, 0.007137]
		Internal vs OCT-C8	0.0052 ± 0.0014***	[0.003437, 0.006997]
		OCT-C8 vs LOCTv3	0.0002 ± 0.0018	[-0.001398, 0.002267]

Table 3. Bootstrap-based statistical comparison of external generalization gaps (Δ AUROC) across dataset configurations. Note: Data are presented as mean $\pm$ standard deviation; Δ AUROC is defined as external minus internal performance; CI = confidence interval. Comparisons include Internal vs LOCTv3, Internal vs OCT-C8, and OCT-C8 vs LOCTv3 across OCT2017, Ravelli-derived, and OCT-revised datasets. Statistical significance is based on FDR-adjusted p-values (Benjamini–Hochberg correction), indicated as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Analysis of AUROC differences further revealed that, although some comparisons reached statistical significance, the magnitude of the observed differences remained

small (ΔAUROC typically < 0.01 across models). This indicates that leakage effects, while detectable, have limited practical impact on ranking-based performance metrics.

Overall, these results demonstrate that statistical significance does not necessarily imply practical significance, particularly for AUROC. In contrast, classification-based metrics such as macro-F1 showed greater sensitivity to dataset differences, highlighting their importance in evaluating clinically meaningful performance changes.

4 Discussion

4.1 Methodological Contribution and Key Findings

This study presents a leakage-aware evaluation framework for OCT classification and enables a systematic investigation of how different forms of data leakage influence model performance. By progressively constructing three dataset configurations with increasing levels of leakage control—OCT2017 (leakage-prone), Ravelli-derived (duplicate-controlled), and OCT-revised (fully leakage-controlled)—we were able to isolate the effects of both file-level and patient-level leakage within a unified experimental setting.

Quantitatively, model performance remained consistently high across all datasets, with macro-AUROC exceeding 0.99 in OCT2017 and Ravelli, and slightly decreasing to approximately 0.992–0.993 in the OCT-revised dataset. In contrast, macro-F1 showed a more substantial reduction, decreasing from roughly 0.96 in OCT2017 to approximately 0.92–0.93 in OCT-revised. The performance difference between OCT2017 and Ravelli was negligible ($\Delta\text{AUROC} < 0.001$), despite the removal of 7,877 duplicated images. However, the transition from Ravelli to OCT-revised resulted in a consistent reduction across all models.

These results demonstrate that, under a 5-fold cross-validation setting, removing duplicated images alone has minimal impact on performance, while enforcing patient-

level separation leads to a measurable decline. At the same time, models trained with stricter leakage control show better generalization stability. Collectively, these findings reveal that patient-level leakage is the primary contributor to performance inflation, while file-level duplication plays a relatively minor role.

4.2 Comparison with Previous Literature

Previous studies on OCT classification have consistently reported near-perfect performance, often exceeding AUROC values of 0.98 or even approaching 1.00 [1–4]. However, such performance levels are comparable to those observed in the OCT2017 dataset in this study, where AUROC values reached approximately 0.996–0.998 across models.

These findings are in line with prior reports using the same dataset. For instance, Kermany et al. [5] reported AUROC values approaching 0.99 on OCT2017, while subsequent studies have reproduced similarly high performance under conventional evaluation settings. Compared to these studies, our results under leakage-prone conditions remain comparable, suggesting that such near-ceiling performance is reproducible but may not reflect true generalization.

Recent work has raised concerns regarding the validity of these results. Tampu et al. [9] demonstrated that data leakage in OCT datasets can significantly inflate classification performance, while Kapoor and Narayanan [15] further highlighted that hidden correlations and dataset leakage are widespread issues in machine learning research. Consistent with these observations, our results show that when stricter leakage control was applied, AUROC decreased slightly by approximately 0.003–0.005, while macro-F1 showed a larger reduction of approximately 0.03–0.04.

Notably, compared to prior studies reporting near-perfect performance, the reduction observed in our leakage-controlled setting suggests that conventional evaluation protocols may overestimate model performance, particularly when patient-level overlap is not explicitly addressed. Despite the relatively modest decrease in AUROC,

the more pronounced decline in macro-F1 indicates that clinically relevant classification performance may be more substantially affected than suggested by ranking-based metrics alone.

4.3 External Validation and Generalization Gap

Furthermore, while external validation is widely used in prior studies, our findings demonstrate that external datasets may not always be independent. Recent studies have emphasized that the reliability of external validation critically depends on the assumption of dataset independence, which is often difficult to verify in practice [7]. For example, Wynants et al. [30] highlighted that improper validation strategies can lead to overly optimistic performance estimates, while Poon et al. [8] discussed the challenges of hidden overlap and shared data sources in publicly available medical datasets.

In our study, evaluation on OCT-C8 produced positive performance differences ($\Delta\text{AUROC} \approx +0.001$ to $+0.003$), suggesting artificially improved generalization. This observation is consistent with prior reports indicating that hidden dataset overlap can lead to misleading conclusions regarding model robustness [33].

In contrast, evaluation on LOCTv3 revealed a markedly different pattern. Models trained on OCT2017 and Ravelli exhibited near-zero or slightly negative differences ($\Delta\text{AUROC} \approx -0.001$ to $+0.001$), whereas models trained on the leakage-controlled OCT-revised dataset showed consistently positive differences ($\Delta\text{AUROC} \approx +0.003$ to $+0.006$). Compared to previous studies that relied solely on external validation without verifying dataset independence, our results suggest that such evaluations may overestimate generalization performance [34].

This pattern indicates that LOCTv3 provides a more realistic estimate of model generalization, while also highlighting that stricter leakage control improves robustness to distributional shifts. These findings align with broader observations in medical AI that robust generalization requires both rigorous dataset curation and

careful validation design.

4.4 Impact of Leakage: Internal vs Leakage-Controlled Settings

The comparison across OCT2017, Ravelli, and OCT-revised datasets provides direct quantitative evidence of how data leakage affects model evaluation. Internal performance remained near ceiling in OCT2017 and Ravelli, with AUROC values consistently above 0.996 and minimal variance across models (standard deviation typically < 0.001). Such compressed performance ranges limit the ability to distinguish between models and obscure meaningful differences. Similar near-ceiling performance has been widely reported in previous OCT classification studies [1–4].

However, recent work has suggested that such limited variability may reflect dataset-specific biases rather than true model capability. Sambasivan et al. [35] highlighted the impact of data-related issues in machine learning pipelines, while Nguyen et al. [36] demonstrated that dataset bias can substantially affect generalization in medical imaging AI. Compared to these studies, our results show that performance variability becomes more apparent when stricter leakage control is applied, revealing differences that are otherwise masked under leakage-prone conditions.

By contrast, the OCT-revised dataset showcased a broader performance distribution, with macro-F1 values decreasing by approximately 3–4% and increasing variability across folds. This observation is consistent with prior findings that removing hidden correlations can restore meaningful performance variation and provide more reliable estimates of model behavior under distributional shift [36,37].

The minimal difference between OCT2017 and Ravelli confirms that duplicate removal alone is insufficient. Instead, the substantial performance shift observed in the OCT-revised dataset indicates that patient-level leakage is the primary factor driving inflated performance in OCT classification tasks. This finding aligns with previous studies suggesting that subject-level dependencies represent a major source of hidden bias in medical imaging datasets [35,36].

4.5 Metric Sensitivity: AUROC vs Macro-F1

A consistent pattern observed in this study is the discrepancy between AUROC and macro-F1. Across all dataset configurations, AUROC differences remained small (typically $\Delta\text{AUROC} < 0.01$), even when statistically significant. In contrast, macro-F1 exhibited larger relative changes, with reductions of approximately 3–4% following leakage removal.

This divergence reflects the inherent properties of these metrics. AUROC, as a ranking-based metric, remains stable under near-ceiling conditions and may fail to capture clinically relevant degradation in classification performance. By comparison, macro-F1, which depends on classification thresholds, is more sensitive to misclassification and class imbalance.

These findings suggest that AUROC alone may underestimate the impact of data leakage, and that classification-based metrics such as macro-F1 provide more informative insights into practical model performance.

4.6 Limitations

This study has several limitations. Although duplicate removal and patient-level separation were systematically implemented to mitigate major sources of data leakage, additional forms of bias may still persist. In particular, near-duplicate images (e.g., adjacent OCT slices) and acquisition-level correlations were not explicitly quantified. Such hidden correlations and dataset-specific biases have been widely recognized as important challenges in medical imaging AI and may lead to overly optimistic performance estimates [38,41].

Furthermore, this study relied on publicly available datasets, which, despite enabling reproducibility and large-scale analysis, may not fully capture the heterogeneity of real-world clinical environments. Prior work has shown that models trained on curated datasets may exhibit reduced performance under real-world distributional

shifts [39,40].

In addition, external validation was conducted using two datasets, one of which (OCT-C8) was later found to contain substantial overlap with the internal dataset. This reflects a broader challenge reported in previous studies, where external datasets may not be fully independent despite being treated as such, potentially leading to misleading estimates of model generalization [38].

Future work incorporating prospectively collected, multi-center datasets and more advanced leakage detection strategies would further improve the robustness and clinical applicability of these findings.

5. Conclusion

In summary, our findings demonstrate that data leakage can substantially bias performance evaluation in OCT classification. While deep learning models often achieve near-perfect performance under conventional evaluation settings, quantitative analysis reveals that such results may not accurately reflect true generalization capability.

The proposed leakage-aware evaluation paradigm offers a more reliable method of assessing model performance. Importantly, the ramifications of this study go beyond OCT classification. Data leakage is a widespread problem in medical imaging and machine learning, particularly in datasets containing repeated measurements, longitudinal scans, or patient-level correlations. Therefore, robust leakage control—including duplication detection, subject-level separation, and verification of external dataset independence—should be regarded as a critical component of evaluation methodology in medical AI.

Future research should concentrate on building more advanced leakage detection algorithms, including clinically obtained datasets, and systematically reporting generalisation gaps alongside traditional performance indicators. Ultimately,

evaluation in medical AI should prioritize robustness and clinical validity over high numerical performance.

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Acknowledgements

Author contributions

TYL and CJJ conceived the study. TYL, HHC, CCC, and CJJ designed the methodology. TYL and HHC implemented the software, performed validation, and conducted the formal analysis. HHC and CCC collected the data and curated the datasets. TYL and HHC drafted the manuscript. CCC managed the project. CJJ provided resources, supervised the study, and acquired funding. All authors reviewed and edited the manuscript and approved the final version.

Data availability

All datasets used in this study are publicly available from the sources cited in the References. OCT2017, the Ravelli-derived dataset, and OCT-C8 are available through Kaggle, and LOCTv3 is available through Mendeley Data. The code and processed data generated during the current study are available from the corresponding author upon reasonable request.

Competing interests

The authors declare no competing interests.

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Figures and legends:

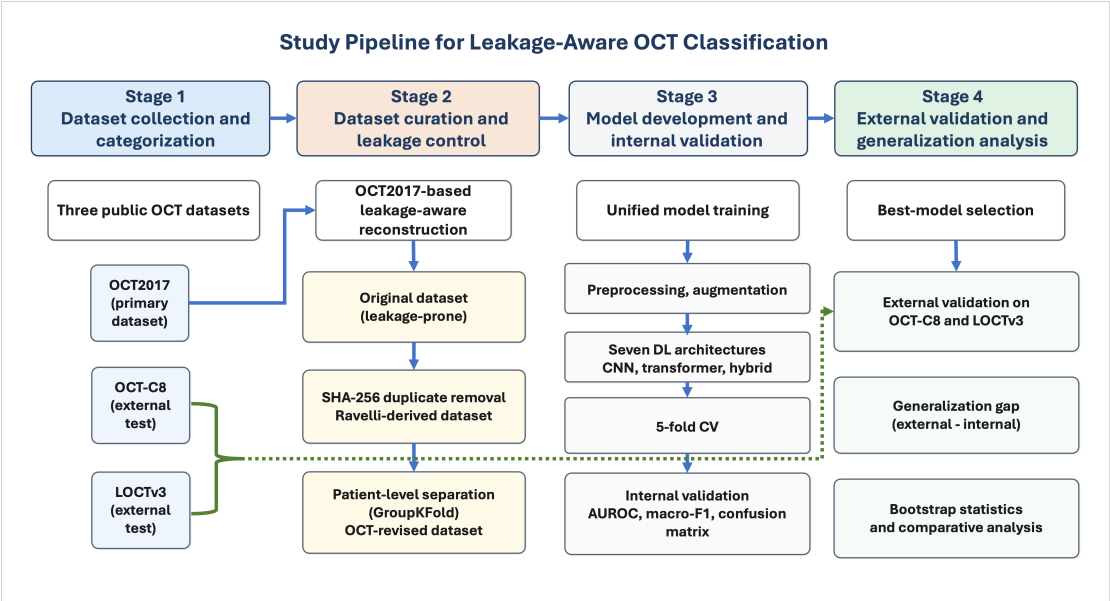


Fig. 1. Study pipeline for leakage-aware OCT classification. The workflow comprised four stages: dataset collection and categorization, dataset curation and leakage control, model development and internal validation, and external validation with generalization analysis. OCT2017 was used as the primary development dataset and was progressively reconstructed into three versions with increasing leakage control: the original leakage-prone dataset, a duplication-controlled dataset (Ravelli-derived), and a fully leakage-controlled dataset with patient-level separation (OCT-revised). Multiple deep learning models were then trained and internally evaluated under a unified protocol. Finally, the best-performing model was externally validated on OCT-C8 and LOCTv3, and generalization gaps were quantified using bootstrap-based statistical analysis.

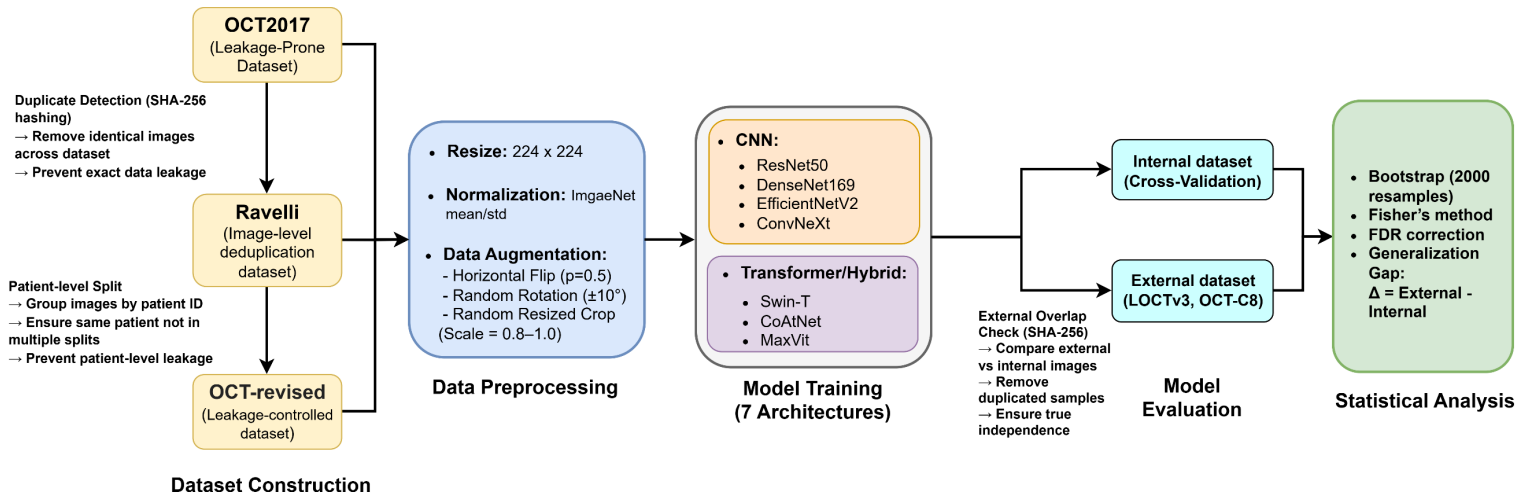


Fig. 2. Leakage-aware experimental framework for OCT classification. The pipeline includes dataset curation with SHA256-based duplicate removal and patient-level partitioning to prevent cross-set contamination, followed by standardized image preprocessing and augmentation. Multiple model architectures, including convolution neural networks and transformer-based models, are trained under a unified protocol. Performance is evaluated on both internal and external datasets, and rigorous statistical analyses are conducted to quantify performance differences and assess the impact of data leakage on model generalization.

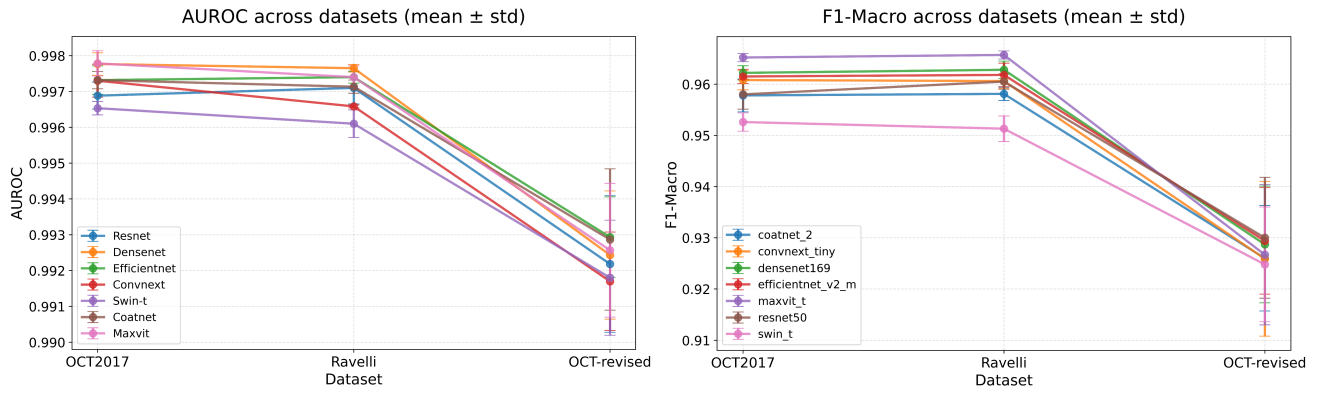


Fig. 3. Internal performance comparison across datasets with different levels of leakage control. The figure shows macro-AUROC (a) and macro-F1 (b) for all models across OCT2017, Ravelli, and OCT-revised datasets (mean $\pm$ standard deviation over five folds). Performance remained consistently high, with minimal differences between OCT2017 and Ravelli. In contrast, a systematic decline was observed in the OCT-revised dataset after stricter leakage control. This reduction was more pronounced in macro-F1 than in AUROC, indicating that threshold-dependent metrics are more sensitive to leakage removal.

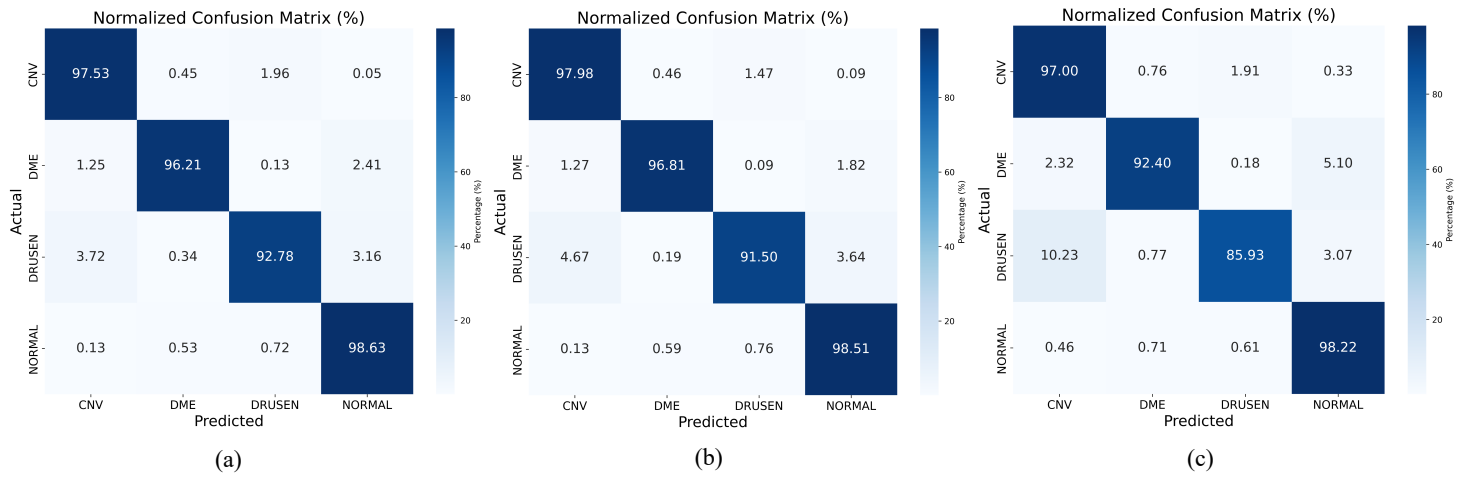


Fig. 4. The figure shows normalized confusion matrices (%) of EfficientNet-V2-M across three dataset configurations. (a): OCT2017; (b): Ravelli; (c): OCT-revised

Distribution of DRUSEN → NORMAL Misclassification Across Folds

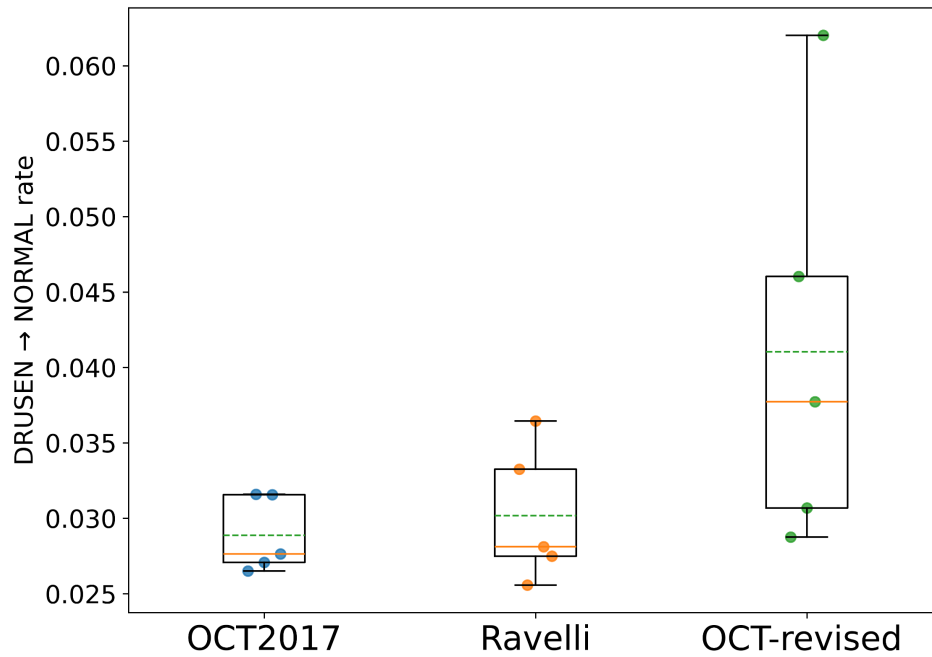


Fig. 5. Distribution of DRUSEN-to-NORMAL misclassification rates across folds for OCT2017, Ravelli, and OCT-revised datasets. In each boxplot, the central line represents the median misclassification rate, the box indicates the interquartile range (IQR; 25th to 75th percentile), and the whiskers extend to the range of observed values excluding outliers. Individual points correspond to results from each fold (5-fold), while the dashed line indicates the mean value.

Tables

Table 1. Class distribution across OCT dataset versions after progressive cleaning.

CLASS	ORIGINAL	SHA256	GKFOLD
CNV	37,455	31,638	31,638
DME	11,598	10,985	10,985
DRUSEN	8,866	7,821	7,821
NORMAL	26,565	26,163	26,163

Table 2. Data leakage analysis across OCT dataset versions after SHA-256-based duplicate removal and GroupKFold-based patient-level separation.

Stage	Images	SHA256	SHA256	SHA256	Patient	Patient	Patient
		Train∩Val	Train∩Test	Val∩Test	Train∩Val	Train∩Test	Val∩Test
Original	84,484	844	2093	225	2147	3262	1813
SHA256	76,607	0	0	0	2121	3230	1783
GKFOLD	76,607	0	0	0	0	0	0

Table 3. Overall performance comparison across dataset configurations for OCT classification.

	Model	AUROC	Recall	Specificity	PPV	NPV	F1 score
OCT2017	coatnet_2	0.9973 $\pm$ 0.0003	0.9615 $\pm$ 0.0015	0.9899 $\pm$ 0.0006	0.9544 $\pm$ 0.0055	0.9891 $\pm$ 0.0010	0.9578 $\pm$ 0.0032
	convnext_tiny	0.9973 $\pm$ 0.0005	0.9635 $\pm$ 0.0021	0.9906 $\pm$ 0.0004	0.9584 $\pm$ 0.0037	0.9900 $\pm$ 0.0007	0.9608 $\pm$ 0.0019
	densenet169	0.9978 $\pm$ 0.0003	0.9632 $\pm$ 0.0012	0.9908 $\pm$ 0.0003	0.9612 $\pm$ 0.0015	0.9906 $\pm$ 0.0003	0.9622 $\pm$ 0.0014
	efficientnet_v2_m	0.9973 $\pm$ 0.0004	0.9630 $\pm$ 0.0021	0.9907 $\pm$ 0.0004	0.9600 $\pm$ 0.0011	0.9903 $\pm$ 0.0003	0.9615 $\pm$ 0.0014
	maxvit_t	0.9978 $\pm$ 0.0003	0.9676 $\pm$ 0.0011	0.9917 $\pm$ 0.0002	0.9629 $\pm$ 0.0014	0.9912 $\pm$ 0.0002	0.9652 $\pm$ 0.0008
	resnet50	0.9969 $\pm$ 0.0004	0.9596 $\pm$ 0.0042	0.9897 $\pm$ 0.0008	0.9565 $\pm$ 0.0029	0.9894 $\pm$ 0.0006	0.9580 $\pm$ 0.0029
	swin_t	0.9965 $\pm$ 0.0002	0.9535 $\pm$ 0.0024	0.9883 $\pm$ 0.0005	0.9518 $\pm$ 0.0021	0.9881 $\pm$ 0.0004	0.9526 $\pm$ 0.0018
Ravelli	coatnet_2	0.9971 $\pm$ 0.0002	0.9600 $\pm$ 0.0008	0.9899 $\pm$ 0.0003	0.9563 $\pm$ 0.0021	0.9896 $\pm$ 0.0004	0.9581 $\pm$ 0.0013
	convnext_tiny	0.9966 $\pm$ 0.0000	0.9609 $\pm$ 0.0012	0.9905 $\pm$ 0.0003	0.9603 $\pm$ 0.0022	0.9904 $\pm$ 0.0005	0.9606 $\pm$ 0.0014
	densenet169	0.9976 $\pm$ 0.0001	0.9634 $\pm$ 0.0017	0.9911 $\pm$ 0.0003	0.9623 $\pm$ 0.0029	0.9910 $\pm$ 0.0004	0.9628 $\pm$ 0.0017
	efficientnet_v2_m	0.9974 $\pm$ 0.0002	0.9631 $\pm$ 0.0020	0.9908 $\pm$ 0.0006	0.9605 $\pm$ 0.0030	0.9906 $\pm$ 0.0006	0.9618 $\pm$ 0.0023
	maxvit_t	0.9974 $\pm$ 0.0003	0.9669 $\pm$ 0.0013	0.9918 $\pm$ 0.0002	0.9645 $\pm$ 0.0011	0.9916 $\pm$ 0.0002	0.9657 $\pm$ 0.0008
	resnet50	0.9971 $\pm$ 0.0004	0.9603 $\pm$ 0.0018	0.9904 $\pm$ 0.0003	0.9607 $\pm$ 0.0016	0.9904 $\pm$ 0.0003	0.9605 $\pm$ 0.0015
	swin_t	0.9961 $\pm$ 0.0004	0.9515 $\pm$ 0.0014	0.9882 $\pm$ 0.0006	0.9511 $\pm$ 0.0039	0.9881 $\pm$ 0.0008	0.9513 $\pm$ 0.0025
OCT-revised	coatnet_2	0.9929 $\pm$ 0.0020	0.9340 $\pm$ 0.0083	0.9820 $\pm$ 0.0023	0.9197 $\pm$ 0.0145	0.9806 $\pm$ 0.0034	0.9260 $\pm$ 0.0103
	convnext_tiny	0.9917 $\pm$ 0.0014	0.9315 $\pm$ 0.0088	0.9820 $\pm$ 0.0035	0.9218 $\pm$ 0.0205	0.9810 $\pm$ 0.0048	0.9259 $\pm$ 0.0151
	densenet169	0.9924 $\pm$ 0.0018	0.9337 $\pm$ 0.0084	0.9826 $\pm$ 0.0026	0.9248 $\pm$ 0.0156	0.9818 $\pm$ 0.0035	0.9287 $\pm$ 0.0114
	efficientnet_v2_m	0.9929 $\pm$ 0.0011	0.9346 $\pm$ 0.0079	0.9828 $\pm$ 0.0024	0.9249 $\pm$ 0.0147	0.9818 $\pm$ 0.0031	0.9294 $\pm$ 0.0104
	maxvit_t	0.9926 $\pm$ 0.0019	0.9342 $\pm$ 0.0140	0.9822 $\pm$ 0.0034	0.9212 $\pm$ 0.0146	0.9810 $\pm$ 0.0037	0.9267 $\pm$ 0.0137
	resnet50	0.9922 $\pm$ 0.0019	0.9337 $\pm$ 0.0099	0.9829 $\pm$ 0.0026	0.9268 $\pm$ 0.0144	0.9821 $\pm$ 0.0032	0.9300 $\pm$ 0.0118
	swin_t	0.9918 $\pm$ 0.0016	0.9293 $\pm$ 0.0085	0.9816 $\pm$ 0.0026	0.9210 $\pm$ 0.0143	0.9807 $\pm$ 0.0033	0.9248 $\pm$ 0.0112

Note: Data are presented as mean $\pm$ standard deviation; AUROC = area under the receiver operating characteristic curve; PPV = positive predictive value; NPV = negative predictive value; F1(M) = macro-averaged F1 score. Results are reported for OCT2017, Ravelli-derived, and OCT-revised, corresponding to progressively increasing levels of leakage control.

